Do Now · retrieval

Arrays in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

For a 3 × 3 array called board , what do GetLength(0) and GetLength(1) return?

Q2

What is the difference between a function and a procedure?

Q3

Convert hexadecimal 2F to denary.

Q4

Why must every value in a C# string[,] array be stored as a string?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

1D · one index — CODE:
string[] students =
{
"Amelia", "Ben", "Charlie", "Daisy"
};

Console.WriteLine(students[2]); — Indexes start at 0, so students[2] is Charlie .

Core idea 2

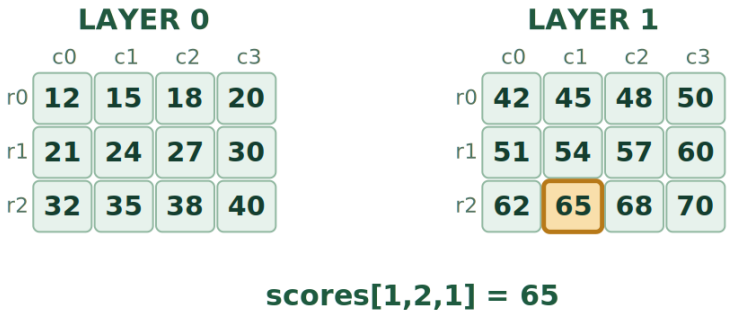
2D · row and column — CODE:
string[,] studentResults =
{
{ "Amelia", "72" },
{ "Ben", "65" },
{ "Charlie", "81" }
};

Console.WriteLine(studentResults[0,0]); — [0,0] means row 0, column 0. It outputs Amelia .

Core idea 3

GetLength() names the dimension — CODE:
studentResults.GetLength(0) // rows
studentResults.GetLength(1) // columns — For the array above, the answers are 3 and 2 .

Key visual · complete concept



scores[1,2,1] = 65

New Learning · applying the ideas

Core idea 4

3D · layer, row, column — CODE:

```
int[,,] scores =
{
{
{12, 15, 18, 20},
{21, 24, 27, 30},
{32, 35, 38, 40}
},
{
{42, 45, 48, 50},
{51, 54, 57, 60},
{62, 65, 68, 70}
}
};
```

— Read the indexes as layer → row → column . A 3D array is easiest to visualise as several 2D tables stacked together.

Vocabulary · use the correct technical terms

Array — A fixed collection of elements accessed using indexes.

Element — One stored value inside an array.

Index — The position number used to access an array element.

Dimension — One direction or level used to locate an element.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Build studentResults

Follow the worked method

1. Rows are students · columns are fields — CODE:
row 0 → Amelia | 72
row 1 → Ben | 65
row 2 → Charlie| 81 — We need 3 rows and 2 columns.

2. Use row, then column — CODE:
Console.WriteLine(studentResults[0,0]);
Console.WriteLine(studentResults[0,1]); — The outputs are Amelia and 72 .

3. Three dimensions need three loops — CODE:

```
for (int layer = 0;
layer < scores.GetLength(0);
layer++)
{
for (int row = 0;
row < scores.GetLength(1);
row++)
{
for (int column = 0;
column < scores.GetLength(2);
column++)
{
Console.WriteLine(
scores[layer,row,column]);
}
}
}
```

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Read the array

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Find 24

Using the supplied scores array, change the Console.WriteLine statement so the program outputs 24 .

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Find the largest value
Use three nested loops to find and display the largest value in the entire 3D array.
Do not assume 48 is the largest. Your algorithm must still work if the values in the array are changed.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

int[,,] data = new int[2,3,4]; declares a three-dimensional array.

Use key vocabulary from memory

1. Explain Array in your own words.

2. Explain Element in your own words.

3. Explain Index in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.